FWDGCS

FWD Ground Control Station

Software User Manual v2.0

Version: 2.0

Document Version: 1.0

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Platform: Windows / Linux / Android

FWD Defence
fwdefence.com

Disclaimer
FWDGCS is designed and manufactured for professional applications. Operators are expected to possess fundamental technical skills. Please handle this product with care.

FWD Defence shall not be held liable for any economic loss or personal injury resulting from improper or unsafe operation of this product.

Unauthorized disassembly or modification is strictly prohibited without prior consent from FWD Defence.

This manual addresses most common usage questions. For additional assistance, please contact FWD Defence support.

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1 Product Overview

Functions:
FWDGCS provides functionalities including flight control, aircraft configuration, UAV data monitoring, mission planning, camera operation, video transmission, and login authentication.

Applications:

- Flight control and operation
- Mission planning and execution
- Data exchange and processing
- System monitoring and maintenance

Supported UAV Types:

- Multirotor UAVs
- Fixed-wing UAVs
- VTOL Fixed-wing UAVs
- Ground vehicles
- Watercraft

1.1 Product Description

Purpose of the Software:
Flight control and mission execution. Data visualization and decision support.

Core Features:

- Flight control
- Mission management and payload control
- Data exchange and processing
- System maintenance and diagnostics
- Login authentication and session management
- Enhanced video streaming with custom controls

Supported Platform:
Windows 10/11, Ubuntu 20.04/22.04, Android 8.0+

What Makes FWDGCS Different:
FWDGCS extends standard QGC with a custom login/authentication system, enhanced video controls, FWD branding (deep red theme #700000, FWD logo), and Navy-specific pre-configurations.

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1.2 User Guide Structure

Introduction:
This user manual provides detailed instructions and descriptions of each functional module of the software.

Intended Audience:
FWDGCS is designed for professional use. Operators are expected to possess basic technical skills and should exercise caution during operation.

> **Note:** This manual addresses most common usage questions. For additional assistance, please contact FWD Defence support.



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2 Installation and Configuration

2.1 System Requirements

Component	Minimum	Recommended
OS (Desktop)	Windows 10 / Ubuntu 20.04	Windows 11 / Ubuntu 22.04
OS (Mobile)	Android 8.0	Android 11+
Processor	Intel Core i5 2.5 GHz	Intel Core i7 3.0 GHz
RAM	8 GB	16 GB
Storage	500 MB free	10 GB (logs + maps)
Graphics	OpenGL 3.3 support	Dedicated GPU with video decode
Display	1280 x 720	1920 x 1080+

2.2 Installation Steps

2.2.1 Windows Installation

1. Download the FWDGCS installer (`FWD_NAVY_GCS-x.x.x-installer.exe`) from the FWD distribution portal
2. Run the installer as Administrator
3. Follow the setup wizard — accept the license, choose installation directory (default: `C:\Program Files\FWDGCS`), select components
4. Click Install
5. Launch FWDGCS from the Start Menu or desktop shortcut

> **Note:** If video is not working, install GStreamer 1.18.6 runtime separately. Configure Windows Firewall to allow FWDGCS for MAVLink UDP/TCP ports.

2.2.2 Linux Installation

1. Download the AppImage: `FWD_NAVY_GCS-x.x.x-x86_64.AppImage`
2. Make it executable: `chmod +x FWD_NAVY_GCS-x.x.x-x86_64.AppImage`
3. Run: `./FWD_NAVY_GCS-x.x.x-x86_64.AppImage`

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2.2.3 Android Installation

1. Download the APK: `FWD_NAVY_GCS-vx.x.x.apk`
2. Enable Install from Unknown Sources in Settings > Security
3. Open the APK file and follow installation prompts
4. Grant permissions: Location, Storage, USB

2.3 Initial Configuration

When launching the software for the first time, a login window will appear. Log in using your account credentials. After successful login, you can view your account information in System Settings.

> **Warning:** To ensure proper operation of the software, please enable all required permissions when prompted.

[Screenshot: Installation wizard]

[Screenshot: Permission request dialog]

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3 Quick Start

3.1 Main Interface Overview

Upon successful login, the main interface is displayed containing the toolbar, status bar, and view selector. The main views are: Fly, Plan, Setup, and Analyze.

3.1.1 Flight Information

The status bar displays GPS status, battery voltage, telemetry signal strength (RSSI), flight mode, and armed status.



3.2 Quick Start Example

The following is an example of video output, drone connection, and quick takeoff.

3.2.1 Camera Output

Step 1: Tap the three dots in the top-right corner.

Step 2: Select the Camera Module.

Step 3: Tap the downward triangle in the camera address bar and select the correct camera address. Alternatively, tap the pencil icon to manually enter the camera address as prompted.

3.2.2 Data Link Connection

Step 1: Tap the three dots in the top-right corner.

Step 2: Set Data Link to UDP, and set the baud rate to 57600 (ensure it matches the baud rate set in the flight controller).

Step 3: Set the connection type, then click Connect. Enter the address as prompted and click Confirm to establish the connection.

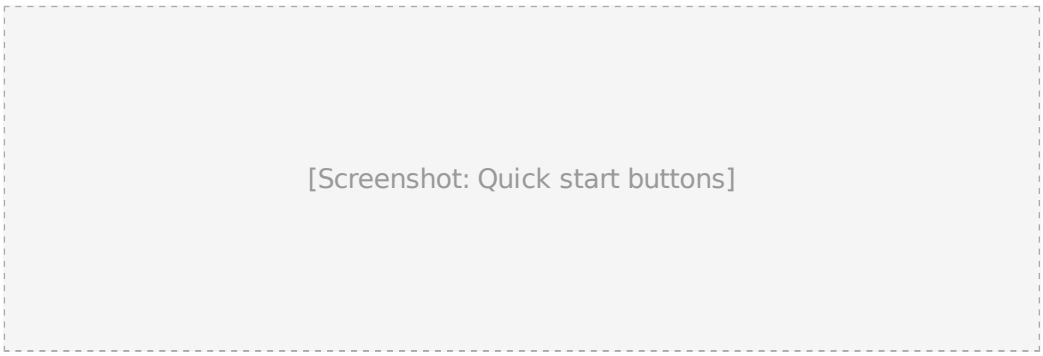
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3.2.3 Shortcut Buttons

One-Touch Takeoff: Tap the one-touch takeoff button, then press and hold as prompted. The drone will climb to 10 meters and hover.

One-Touch Landing: Tap the one-touch landing button, and the drone will land at its current position.

One-Touch Return: Tap the one-touch return button, and the drone will climb to the return altitude and fly back to the takeoff point.



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4 Core Function Instructions

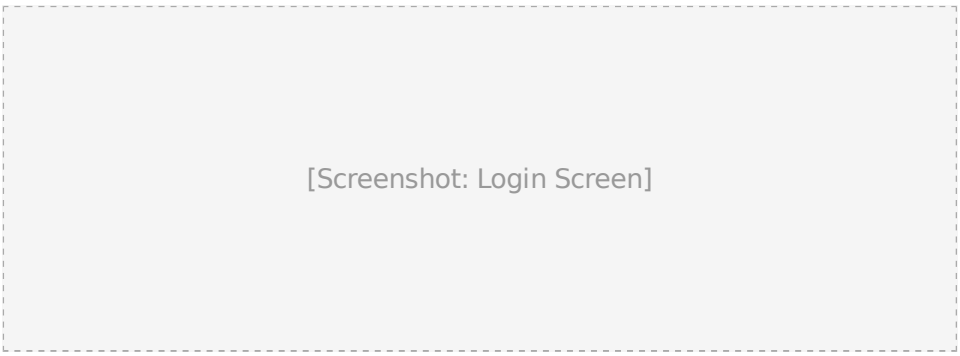
The core functions include: Login Module, Fly View Module, Video System Module, Mission Planning Module, and Vehicle Setup Module.

4.1 Login Module

4.1.1 Login Screen Overview

When FWDGCS starts, the login screen is displayed. Features include:

- FWD Logo displayed prominently at the top
- Title: "GROUND CONTROL STATION" with "FWDGCS" subtitle
- Red gradient theme (#700000)
- Login form with username, password, Remember Me checkbox, and Login button
- Register link to toggle to registration form



4.1.2 Login Procedure

1. Enter your Username and Password
2. Check Remember Me to retain credentials on this device
3. Click Login
4. On successful authentication, the main application window loads

4.1.3 Registration

If you do not have an account, click the Register link and fill in the required fields: Full Name, Username, Email Address, Password, Confirm Password. Click Register to submit.

> **Note:** Registration functionality may require server-side setup. Contact your system administrator if registration is unavailable.

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4.1.4 Session Management

- Your session remains active until you log out or close the application
- Remember Me stores a token locally for automatic login on subsequent launches
- Sessions have a configurable timeout period after which re-authentication is required

4.1.5 Logout

1. Click the profile icon on the main toolbar
2. Select Log Out
3. Confirm the action when prompted
4. The application checks for active vehicle connections before logging out
5. You are returned to the login screen

> **Danger:** Ensure your vehicle is safely landed and disarmed before logging out, as the connection will be terminated.

4.1.6 Login Troubleshooting

Issue	Solution
Invalid credentials	Check username/password; ensure Caps Lock is off
Server unreachable	Check network connectivity; contact administrator
Session expired	Re-enter credentials
Account locked	Wait 15 minutes or contact administrator

4.2 Fly View Module

4.2.1 Main Menu

The Fly View is the primary interface for in-flight operations, consisting of:

- Primary Flight Display (PFD) — artificial horizon, attitude, heading
- Map Panel — vehicle position, waypoints, obstacles
- Video Panel — live camera feed with overlay
- Telemetry Strip — real-time vehicle data
- Action Buttons — Arm, Disarm, Takeoff, Land, RTL, Pause

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4.2.2 Connection Method

Switch the connection type within the flight controller module to the desired mode. Click Connect and follow the on-screen instructions to establish the connection.

Method	Typical Use Case
Serial (UART)	Telemetry radio (SiK, Holybro, etc.)
USB	Direct connection to autopilot
UDP	WiFi bridge, companion computer
TCP	Remote link, relay server

4.2.3 Artificial Horizon / Attitude Indicator

The PFD shows pitch, roll, heading, groundspeed, altitude, and vertical speed.

4.2.4 Telemetry Data Display

Parameter	Description
Altitude	Height above home position (m/ft)
Groundspeed	Speed over ground (m/s / knots)
Airspeed	Speed through air, fixed-wing only (m/s / knots)
Heading	Compass heading (0–360°)
Battery Voltage	Current battery voltage (V)
Battery Remaining	Estimated remaining charge (%)
GPS Satellites	Number of satellites locked
Flight Time	Time since arming (HH:MM:SS)
Link Quality	Radio signal quality (%)

4.2.5 Map Interaction

- Pan: Click and drag

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- Zoom: Scroll wheel, pinch (touch), or +/- buttons
- Centre on Vehicle: Double-click vehicle icon or press C
- Map Layers: Satellite, street, hybrid, or terrain
- Vehicle Trail: Recent path with altitude-colour-coded dots
- Geofence overlay display

4.2.6 In-Flight Controls

Button	Action
Arm	Arms motors — requires pre-arm checks
Disarm	Disarms motors
Takeoff	Autonomous takeoff to specified altitude
Land	Land at current position
RTL	Return to Launch / Home
Pause	Pause mission, hold position

[Screenshot: Fly View with in-flight controls]

4.2.7 Flight Controller Module

4.2.7.1 Flight Settings

Includes speed limits, altitude limits, and failsafe behavior. After entering parameters, click Send to save.

Quick Setting	ArduPilot Parameter
Throttle Failsafe	FS_THR_ENABLE
Follow Speed	WPNAV_SPEED
Return-to-Home Altitude	RTL_ALT
Return-to-Home Speed	RTL_SPEED
Loiter Acceleration Limit	LOIT_ACC_MAX
Maximum Braking Acceleration	LOIT_BRK_ACCEL

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4.2.7.2 Voltage

Setting	ArduPilot Parameter
Low Voltage	BATT_LOW_VOLT
Critical Low Voltage	BATT_CRT_VOLT
Low Voltage Action	BATT_FS_LOW_ACT
Critical Low Voltage Action	BATT_FS_CRT_ACT
Arming Voltage	BATT_ARM_VOLT

4.2.7.3 Accelerometer Menu

Displays accelerometer data and IMU temperature. Provides options for accelerometer calibration and flight controller levelling.

Accelerometer Calibration: Follow instructions to place the drone in six orientations as shown. After completion, restart the drone.

Horizontal Calibration: Place the drone on a flat, level surface and ensure it remains still. Click Horizontal Calibration.

[Screenshot: Accelerometer calibration]

4.2.7.4 Compass Menu

Do not calibrate in areas with strong magnetic fields or near large metal objects. Avoid carrying ferromagnetic materials during calibration.

Steps:

1. Open the compass calibration interface
2. Click Calibrate
3. Lift the drone and rotate in different directions — ensure each side (front, back, left, right, top, bottom) points downward for a few seconds
4. Restart the drone

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4.2.7.5 Remote Control Calibration

Stick Centering: Before calibration, ensure all sticks are centred, confirm correct polarity, then click Calibrate.

Safety Tips: Remove all propellers before calibration. Click Confirm.

Channel Data Logging: Gradually move joysticks and switches to their maximum and minimum positions in sequence.

Calibration Complete: Click Complete to finish. The software displays recorded maximum and minimum values for each channel.

4.2.7.6 Parameter Table

Includes functions for importing, exporting, searching, modifying, refreshing, resetting, and rebooting the flight controller.

Parameter Search: Enter the parameter name and click search.

Parameter Modification: Double-check parameters before modification. Enter the desired value and click Write.

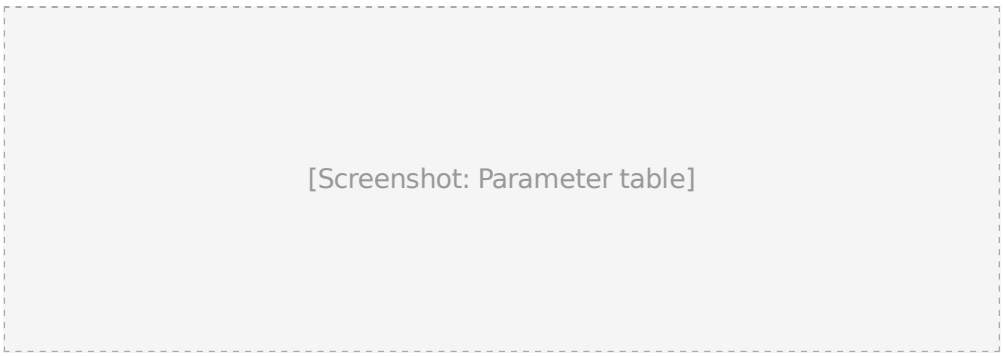
Parameter Export: Click Export Parameters to save all parameters to the remote controller.

Parameter Import: Click Import Parameters and follow prompts to import parameters to the flight controller.

Parameter Reset: Click Reset to restore all parameters to default settings.

Flight Controller Restart: Click Restart Flight Controller to remotely reboot. Must be in unlocked state.

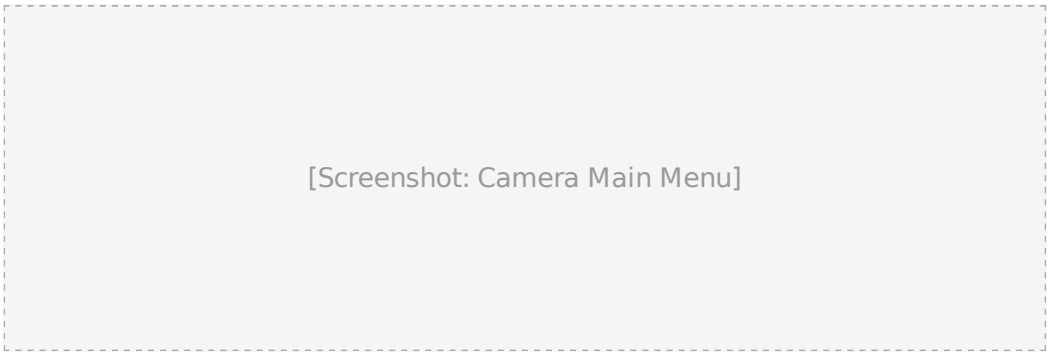
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4.3 Video System Module

4.3.1 Camera Main Menu

Mainly includes camera stream address selection, camera version information, and various control buttons.



4.3.2 Video Stream Address and Version Information

Main Stream / Sub Stream: Set the camera source and parameters.

Address Settings: Configure camera source, select default or manually input RTSP address, disable image display, etc.

> **Note:** When manually entering the address, ensure punctuation marks are in English mode.

Version Information:

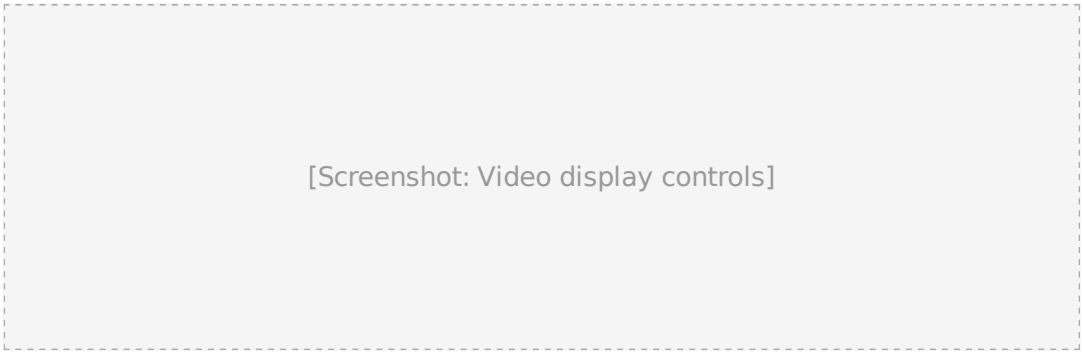
- Camera Firmware Version
- Gimbal Firmware Version
- Zoom Firmware Version (optical zoom cameras only)

4.3.3 Camera Display Operations

- Click minimize button to hide the display
- Drag the arrow at the top of the display to zoom in or out

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- Single-click the small window display to expand to full screen
- After minimizing, click the camera button to restore display
- Long press anywhere and drag the ball to rotate the camera
- Single-click anywhere to auto-focus (zoom cameras)



4.3.4 Gimbal Camera Settings

Set the relevant parameters for the gimbal and camera.

Stream Resolution: Supports HD (720p) and Ultra-HD (1080p).

Auto Record on Startup: Enable/disable automatic video recording to TF card.

Grid Lines: Overlay grid lines and diagonals for compositional assistance.

Video Output Interface: HDMI, CVBS, or Off.

Gimbal Work Modes:

- **Lock Mode:** Gimbal does not follow drone horizontal movement
- **Follow Mode:** Gimbal follows drone horizontal rotation
- **FPV Mode:** Gimbal follows drone roll direction
- **AI Tracking Mode:** AI tracking module connected



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4.3.5 Camera Control

4.3.5.1 Camera Zoom

Control the zoom camera to zoom in and out. Single click = one-time operation; long press = continuous operation.

4.3.5.2 Manual Focus

Manually adjust camera focus. Single click = one-time operation; long press = continuous operation.

4.3.5.3 Photo and Video Recording

Photo: Single click captures a photo. Success message appears on screen.

Video: Click square button to start recording (turns red circle). Recording time and resolution appear in title bar. Click again to stop.

4.3.5.4 Camera Screen Recording

Record the camera screen on the ground station and save locally. Click button to start (highlights). Click again to stop.

4.3.5.5 Gimbal Centering / Lens Down

- Yaw and pitch centering: automatically centres both axes
- Yaw centering + one-touch lens down
- Yaw centering only
- One-touch lens down only

[Screenshot: Camera control buttons]

4.3.5.6 Video Transmission Settings

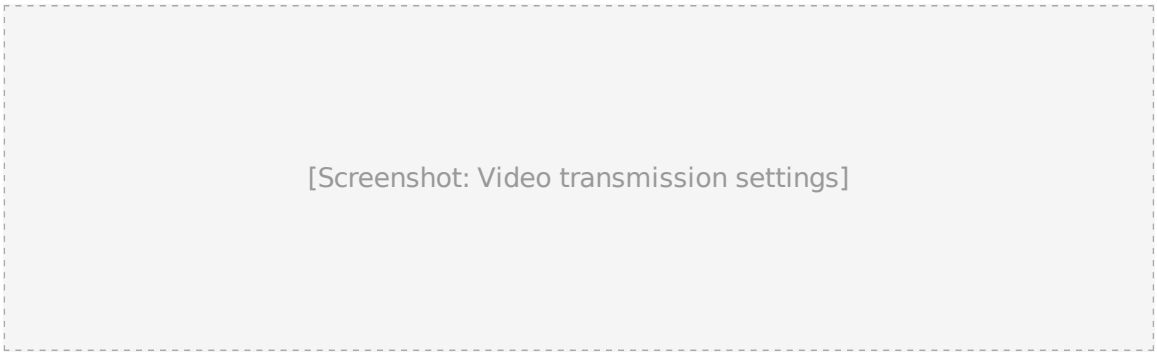
Video Transmission Mode: Change the bitrate mode.

Downlink Bandwidth: Switch maximum bandwidth for video downlink.

Operating Frequency Band: Manually switch frequency band.

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Adaptive Wireless Channel: In complex electromagnetic environments, enable for automatic channel selection. When disabled, manually select channels 1–32.



4.4 Mission Planning Module

4.4.1 Main Menu of the Waypoint Module

- **View and Plan Waypoints:** View waypoint library, plan standard and mapping missions
- **Start/Pause Mission:** Start mission execution. During execution, click to pause (multirotor hovers, fixed-wing loiters, rover holds). Click again to resume
- **Clear Mission:** Clears mission from interface and flight controller. Confirmation required during execution
- **Exit Mission:** Exit current mission (multirotor hovers, fixed-wing loiters, rover holds)
- **Download Mission:** Download mission stored in flight controller
- **Locate Aircraft:** Centre aircraft on map
- **Map Type:** Switch between street and satellite map



4.4.2 Mission Library

Stores planned and saved missions for easy viewing and modification.

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4.4.3 Waypoint Mission

Each waypoint's location must be manually selected. Altitude, speed, and action can be adjusted. The completion action can also be modified.

4.4.3.1 Create Mission

Click Mission Planning button, select Create Waypoint Mission.

4.4.3.2 Waypoint Flight

Assign a name to the mission. Supports switching altitude modes:

- **Absolute Altitude:** Height relative to mean sea level (MSL)
- **Relative Altitude:** Height relative to takeoff point

4.4.3.3 Route Settings

- **Safe Takeoff Altitude:** Altitude to reach before first waypoint
- **Speed:** Horizontal speed during waypoint flight
- **Route Altitude:** Flight altitude during mission
- **Completion Action:** Return to Home, Landing, or Hover

4.4.3.4 Waypoint Settings

- **Speed:** Flying speed after reaching waypoint
- **Height:** Required altitude before reaching waypoint
- **Waypoint Action:** Actions executed after reaching waypoint
- **Hover:** Hover at waypoint for set duration
- **Latitude/Longitude Fine-tuning**

4.4.3.5 Task Execution

1. After planning, click Save
2. Click Start to upload flight route
3. Confirm flight checklist
4. Check route information (speed, altitude range, completion actions)
5. Wait for route upload to complete
6. Unlock the drone and click Execute Task

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4.4.3.6 Jump to Current

Click a specific waypoint, then click Jump to Current. The drone flies directly to the selected waypoint.

> **Note:** If clicking has no effect, download the flight route first, then select a waypoint to execute the jump action.

4.4.4 Mapping Route

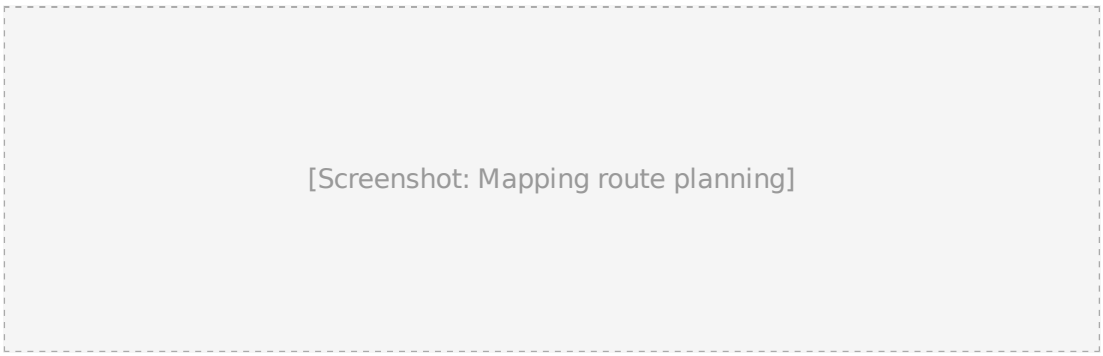
Plan mapping routes by manually selecting boundary points. Modify height, speed, and completion actions.

4.4.4.1 Route Planning

Click Route Planning, select Create Mapping Route.

4.4.4.2 Mapping Route Settings

- **Select Camera:** Choose camera for mapping tasks
- **GSD:** Ground Sample Distance
- **Altitude Mode:** Absolute or Relative
- **Route Speed/Height**
- **Completion Action**
- **Advanced Settings:** Heading overlap, sidelap, external margin



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4.5 Vehicle Setup Module

4.5.1 Main Menu

The Setup View provides a step-by-step wizard for configuring the vehicle.

4.5.2 Firmware

1. Connect the autopilot via USB
2. Go to Setup > Firmware
3. Current firmware version is displayed
4. Select firmware type (ArduPilot or PX4)
5. Click Flash Firmware

4.5.3 Airframe

1. Go to Setup > Airframe
2. Browse or search for your airframe
3. Select and click Apply and Restart

4.5.4 Sensor Calibration

Sensor	Procedure
Accelerometer	Place in 6 orientations as prompted
Magnetometer	Rotate in figure-8 pattern
Gyroscope	Place on level surface, click Calibrate
Level/Horizon	Place in normal flight attitude on level surface
Airspeed	Ensure no wind in pitot tube, click Calibrate

4.5.5 Radio Control Calibration

1. Ensure RC transmitter is bound and powered
2. Go to Setup > Radio
3. Click Calibrate
4. Move each stick to its extremes as instructed
5. Verify channel mapping
6. Set Failsafe values

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4.5.6 Flight Modes

Assign flight modes to RC channel positions. Common modes: Stabilize, Altitude Hold, Loiter, Auto, RTL, Land.

4.5.7 Power & Battery

Configure battery monitor, voltage divider, amps per volt, battery capacity, and low battery thresholds.

[Screenshot: Vehicle setup wizard]

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5 Settings and Preferences

Setting	Description
Language	UI language selection
Units	Metric or Imperial
Map Provider	Google, Bing, Mapbox, OpenStreetMap
Map Type	Satellite, Street, Hybrid, Terrain
Video Source	UDP, RTSP, TCP, or local
Video Decode	Software or hardware
MAVLink Version	1 or 2
Telemetry Logging	Enable/disable .tlog recording
Logging Level	Debug, Info, Warn, Error
Offline Maps	Download regions for offline use

5.1 Software Version Information

In the software settings menu, under About, you can find the current version number of the software.

[Screenshot: Settings screen]

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6 Safety Parameters

Arming Safety Checks:

- GPS Lock — sufficient satellites and 3D fix
- Battery Voltage — within acceptable range
- Accelerometer — calibration valid
- Magnetometer — calibration valid
- Gyroscope — calibration valid
- RC Signal — valid signal received
- Throttle — at minimum

Failsafe Actions:

Action	Behaviour
Continue	Log the event only
Loiter	Hold current position
RTL	Return to home and land
Land	Land immediately
Terminate	Kill motors immediately (emergency only)

Failsafe Triggers:

Failsafe	Recommended Action
RC Loss (no signal >5s)	RTL or Land
Telemetry Loss (>10s)	RTL
Low Battery Stage 1	RTL
Low Battery Stage 2	Land
Geofence Breach	RTL

6.1 RTL Settings

Parameter	Description
RTL Altitude	Altitude to climb before returning (recommended: 50–100m)
RTL Climb	Climb to RTL altitude first (enabled/disabled)

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RTL Loiter Time	Time to loiter before landing (0–30s)
RTL Descend Rate	Vertical speed during landing (0.5–2 m/s)

6.2 Emergency Procedures

Situation	Action
Loss of control	Engage Kill Switch
Flyaway	Engage RTL immediately
Battery critical	Allow automatic land; do not override
Geofence breach	Monitor; vehicle will execute failsafe
Video loss	Continue mission; video is advisory only



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7 Data Persistence and Logs

Data Type	Windows	Linux
Configuration	%APPDATA%/FWDGCS/	~/.config/FWDGCS/
Telemetry Logs	Documents/FWDGCS/Logs/	~/Documents/FWDGCS/Logs/
Missions	Documents/FWDGCS/Missions/	~/Documents/FWDGCS/Missions/
Screenshots	Pictures/FWDGCS/	~/Pictures/FWDGCS/
Recordings	Videos/FWDGCS/	~/Videos/FWDGCS/

Telemetry Logs: FWDGCS automatically logs all MAVLink telemetry while connected. Format: .tlog. One file per session. Naming: YYYY-MM-DD_HH-MM-SS_vehiclename.tlog.

Viewing Logs: Switch to Analyze View (Ctrl+4), click Open Log, select a .tlog file. Use playback controls (Play, Pause, Speed, Timeline).

Crash Reports: If FWDGCS crashes, a report dialog appears on next launch. Click Send Report to transmit anonymized crash data.

[Screenshot: Analyze view with telemetry log]

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8 Troubleshooting

8.1 Common Errors

8.1.1 Unable to Connect to Data Link

1. Ensure the air unit is properly connected to your flight controller
2. Check wiring: TX/RX properly crossed? Data Link configuration correct?
3. In the app, check Link Information — Data Link Downlink should be > 0
4. Check Data Transmission Settings: connection method and baud rate correct?
5. Try switching data cable to TELEM 1 or TELEM 2 port
6. Are both ground station and sky end running latest firmware?
7. If using wireless hotspot for UDP, disable computer Ethernet

8.1.2 Unable to Display Image

1. **Check connection:** Ground and sky end properly synced (green status indicator)? Camera correctly connected to sky end?
2. **Check software settings:** Camera address field correctly set in app?
3. **Check network:** If using handheld ground station, is Ethernet enabled in Android system settings?

8.1.3 Video Issues

Issue	Solution
No video (black screen)	Verify source type/port/URL; check camera is powered
Video freezing	Reduce resolution; check network bandwidth
GStreamer errors	Install all GStreamer plugins (bad, ugly, good)
High CPU usage	Enable hardware decoding; reduce resolution

8.1.4 Connection Issues

Issue	Solution
No heartbeat received	Check baud rate; verify autopilot power; check cable
Serial port not listed	Install USB-to-serial driver; check udev rules (Linux)
Radio not communicating	Verify radio pair; check power; test with another GCS

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UDP/TCP connection refused	Check IP/port; disable firewall temporarily

8.2 Contact Support

Please contact FWD Defence for the latest updates and support.

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9 Appendix

9.1 Glossary

Term	Definition
AHRS	Attitude and Heading Reference System
APM	ArduPilot Mega — open-source autopilot firmware
ESC	Electronic Speed Controller
Failsafe	Automatic behaviour when parameter exceeds safe limits
GCS	Ground Control Station
HDOP	Horizontal Dilution of Precision — GPS accuracy measure
IMU	Inertial Measurement Unit
MAVLink	Micro Air Vehicle Link — communication protocol
PFD	Primary Flight Display
PWM	Pulse Width Modulation
RTK	Real-Time Kinematic — high-precision GPS
RTL	Return to Launch
VTOL	Vertical TakeOff and Landing

9.2 Keyboard Shortcuts

Shortcut	Function
Ctrl+1	Fly View
Ctrl+2	Plan View
Ctrl+3	Setup View
Ctrl+4	Analyze View
Ctrl+F / F11	Toggle fullscreen
Ctrl+N	New plan
Ctrl+O	Open plan file
Ctrl+S	Save plan

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Ctrl+Q	Quit
Ctrl+,	Settings
I	Arm
O	Disarm
Space	RTL (with confirmation)
R	Toggle recording
S	Snapshot
M	Mute/unmute
Delete	Delete selected waypoint
Home	Centre map on vehicle

9.3 Parameter Mapping

ArduCopter 4.7.0 Firmware:

Quick Setting	Parameter
Throttle Fail-safe	FS_THR_ENABLE
Follow Mode Speed	WPNAV_SPEED
RTL Altitude	RTL_ALT
RTL Speed	RTL_SPEED
Loiter Acceleration	LOIT_ACC_MAX
Max Brake Speed	LOIT_BRK_ACCEL
Low Voltage	BATT_LOW_VOLT
Critical Low Voltage	BATT_CRT_VOLT
Low Voltage Behaviour	BATT_FS_LOW_ACT
Critical Low Voltage Behaviour	BATT_FS_CRT_ACT
Arm Voltage	BATT_ARM_VOLT

ArduPlane 4.7.0 Firmware:

Quick Setting	Parameter
Throttle Fail-safe	FS_LONG_ACTN
Cruise Airspeed	AIRSPEED_CRUISE
Max Airspeed	AIRSPEED_MAX

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Min Airspeed	AIRSPEED_MIN
RTL Altitude	RTL_ALTITUDE
Low Voltage	BATT_LOW_VOLT
Critical Low Voltage	BATT_CRT_VOLT
Arm Voltage	BATT_ARM_VOLT

9.4 Version Update Log

Version	Date	Updates
2.0	May 2026	Enhanced video controls; custom login system; FWD branding; safety parameters UI; code cleanup
1.0	Initial	Navy GCS baseline based on upstream QGC

9.5 After-Sales and Warranty

Please contact FWD Defence for after-sales and warranty information.

> **Note:** For additional assistance, please contact FWD Defence support.

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